DOKUZ EYLUL UNIVERSTY

ENGINEERING FACULTY

DEPARTMENT OF COMPUTER ENGINEERING

CME 2210 OBJECT ORIENTED ANALYSIS AND DESIGN

PHASE 2

FACTORY MANAGEMENT SYSTEM

SYSTEM DESIGN REPORT

by

Atakan Kahraman 2022510066

Emre Altundal 2022510092

Serhat Aydın 2023510130

Lecturer

Prof. Dr. Semih Utku

30.04.2025

IZMIR

Contents

1. System Architecture
2. Class Diagram
3. Sequence Diagram
4. Statechart Diagram
5. Activity Diagram
6. Overall System Design Description

System Architecture

The Factory Management System is designed with a modular, scalable, and maintainable structure based on object-oriented programming principles. The system consists of three main layers: Presentation Layer, Application Layer, and Data Layer. Each of these layers contains specific modules that interact with each other through pre-defined rules.

1. Presentation Layer

Displays different interfaces on the user type;

Manager: Shift management, performance review, stock control, and financial operations.

Employee: Viewing salary information, insurance status, work shifts, and performance scores.

Customer: Placing orders, tracking order status, accessing product info, and providing feedback.

2. Application Layer

Authentication Module: Handles login, verifies the user's role, and grants access accordingly.

Employee Management Module: Manages employee data, leave requests, and performance tracking.

Production and Shift Module: Tracks production processes and plans employee shifts.

Order Management Module: Allows customers to place orders, checks stock, and updates order status.

Finance and Payroll Module: Manages salary calculations, payment records, and insurance data.

3. Data Layer

A txt/csv based database will be used.

Stores structured data such as users, products, orders, salaries, shifts, and performance records.

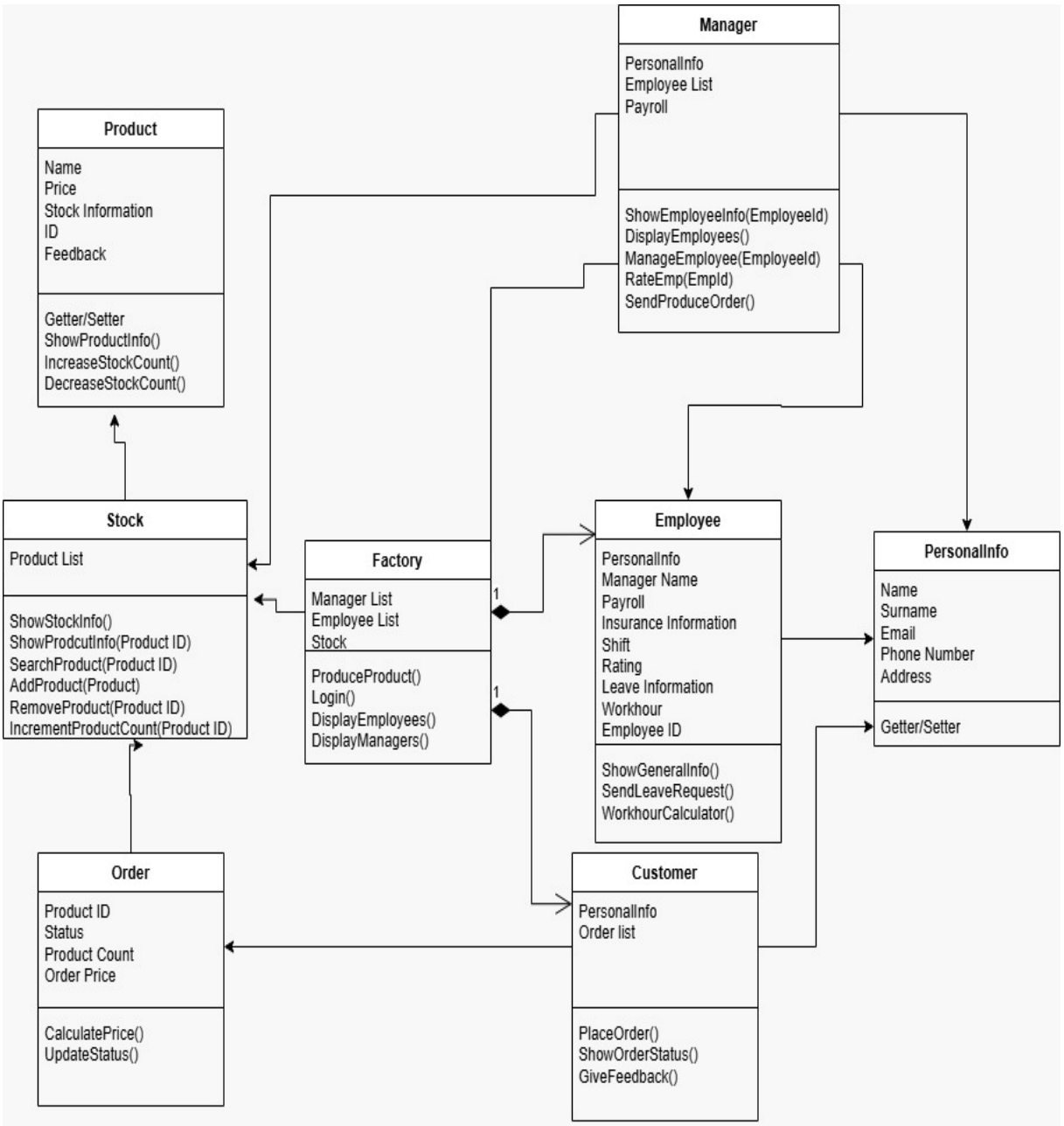
Layer Communication

When a user performs an action, a request is sent to the relevant module and a response is returned. (Example: An employee logs in → the system authenticates the user → they are directed to role-specific pages.)

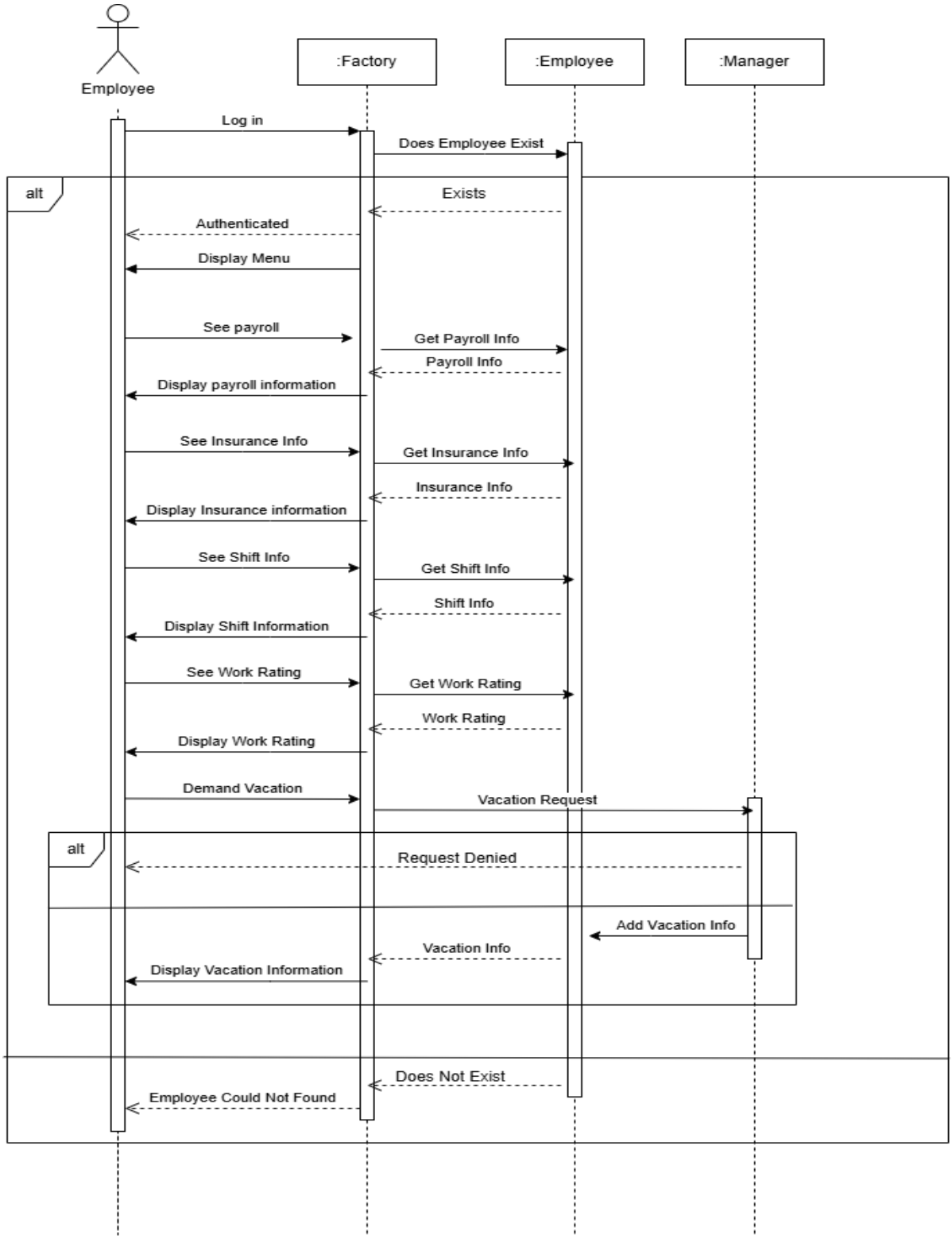
Security

Role-Based Access Control is implemented. Users can only access data and functions relevant to their roles. Also private data is stored securely using abstraction methods such as encapsulation.

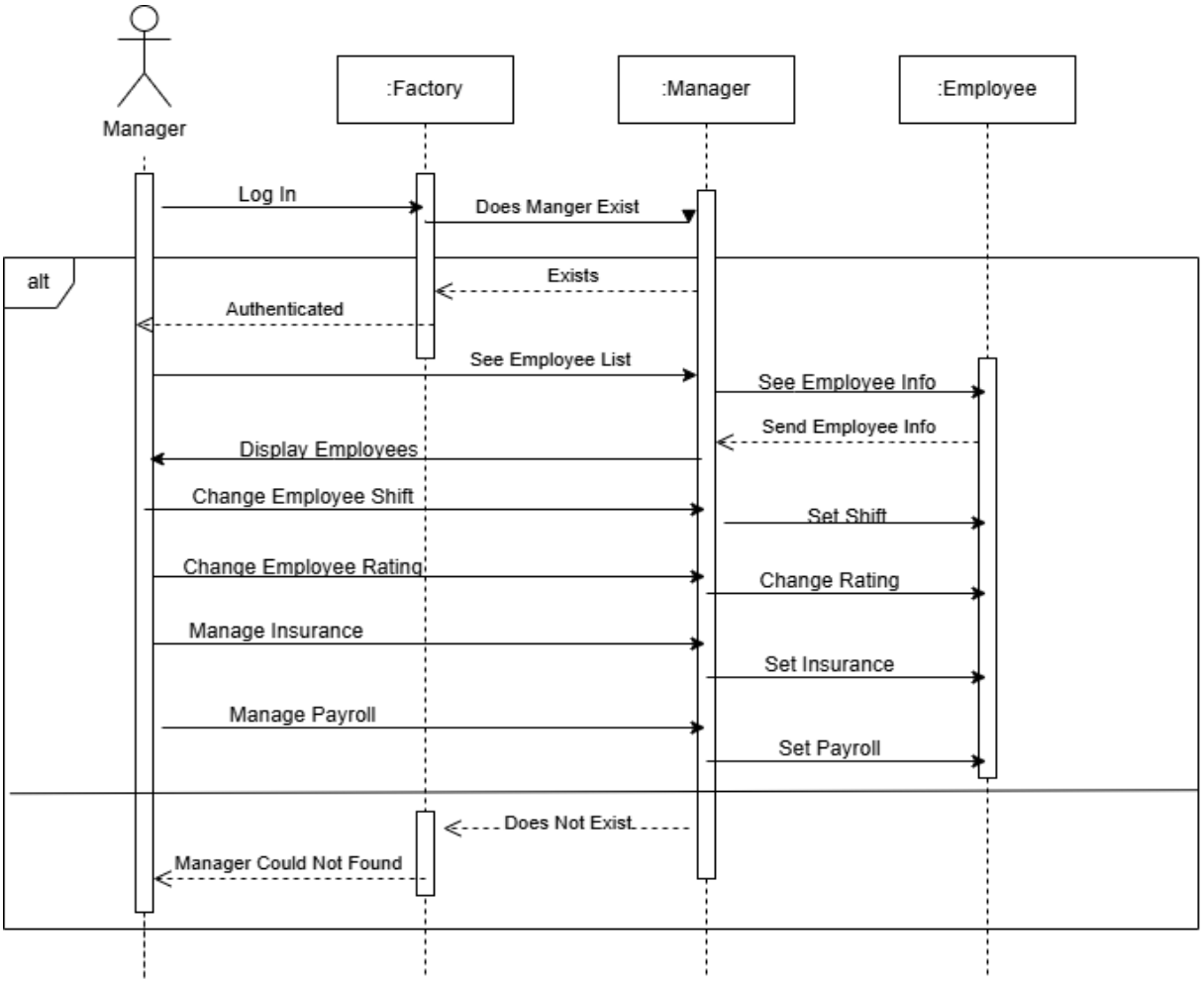
Class Diagram

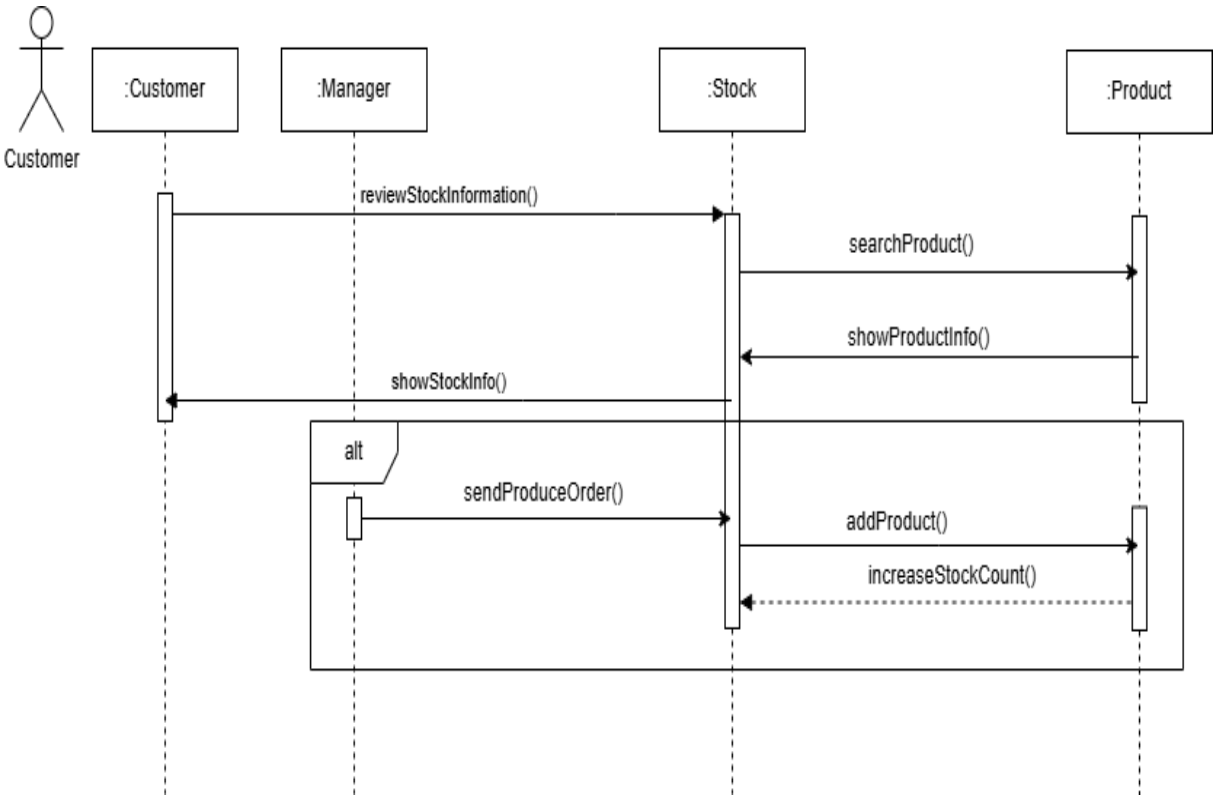


Sequence Diagram 1

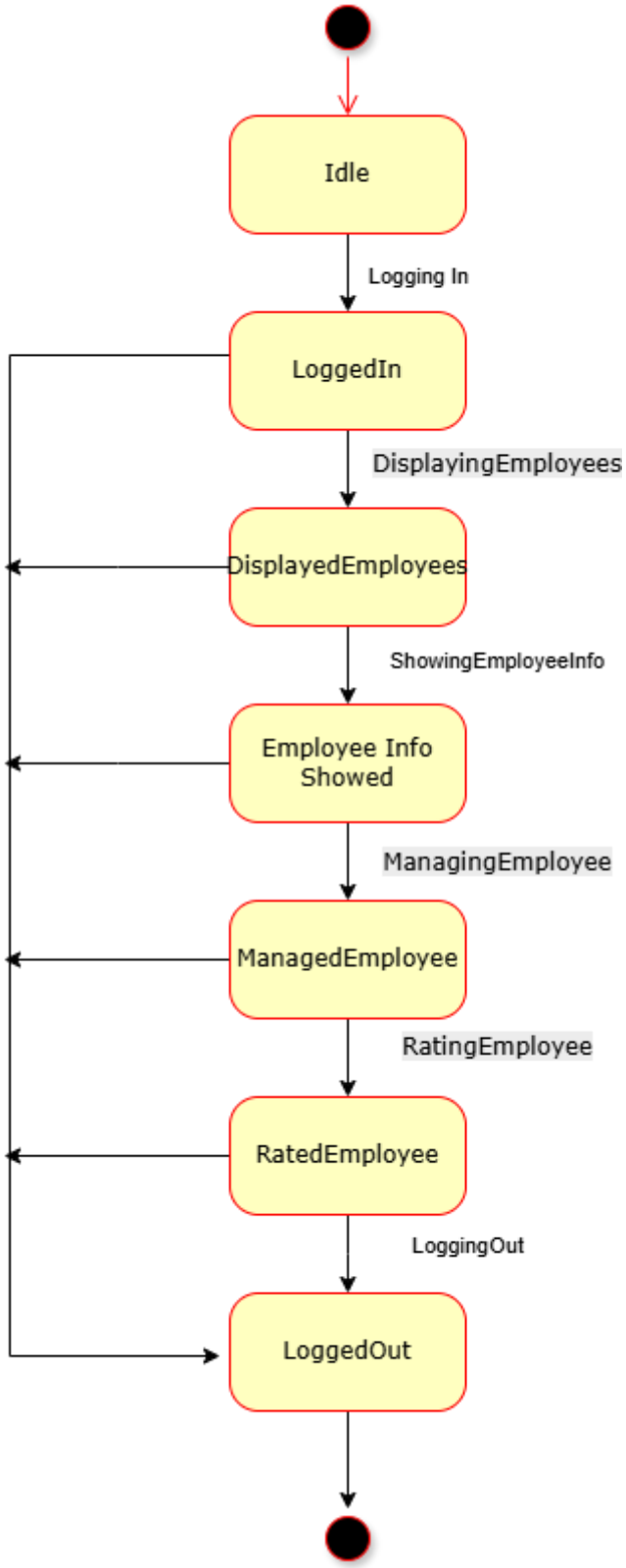


Sequence Diagram 2-3

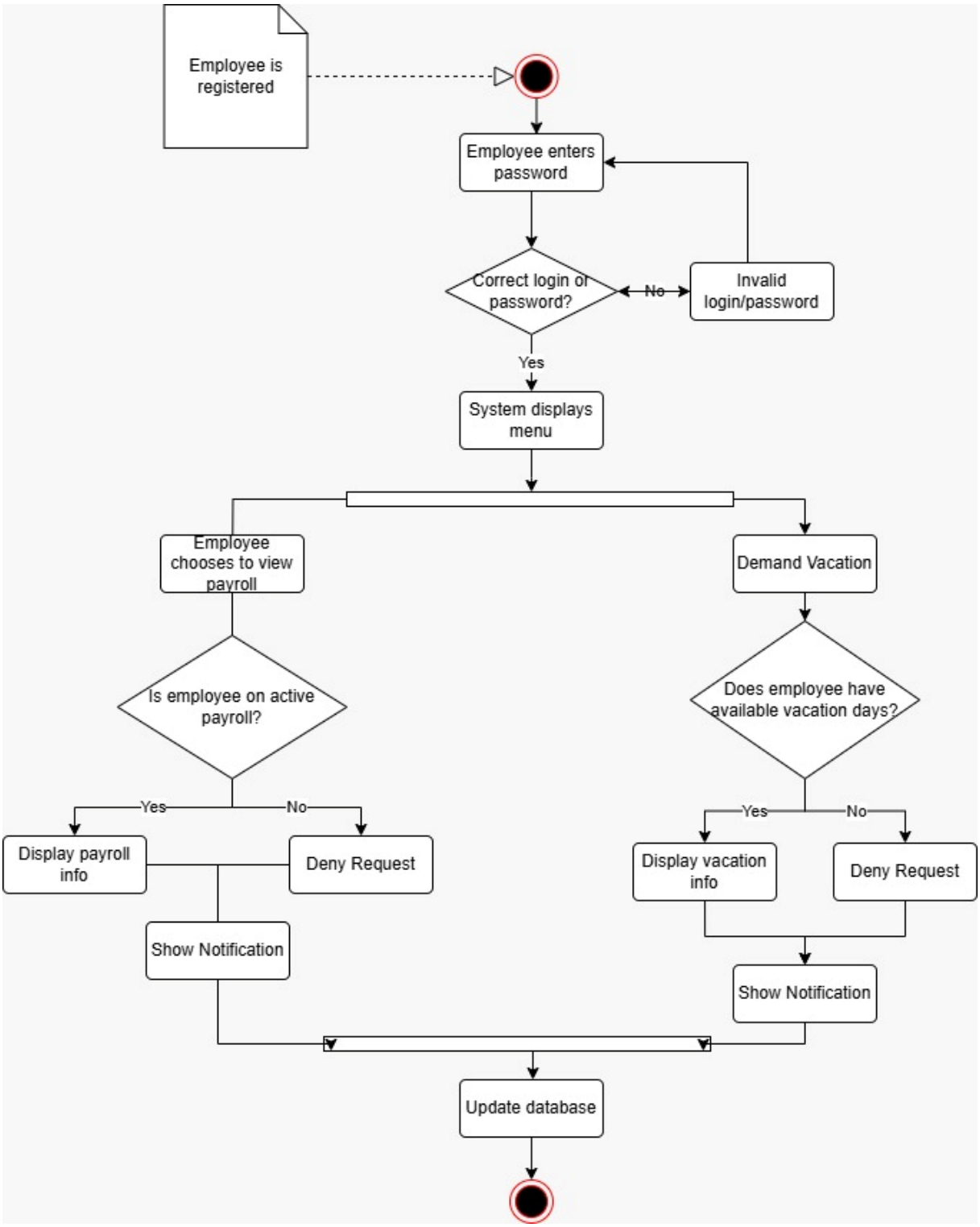




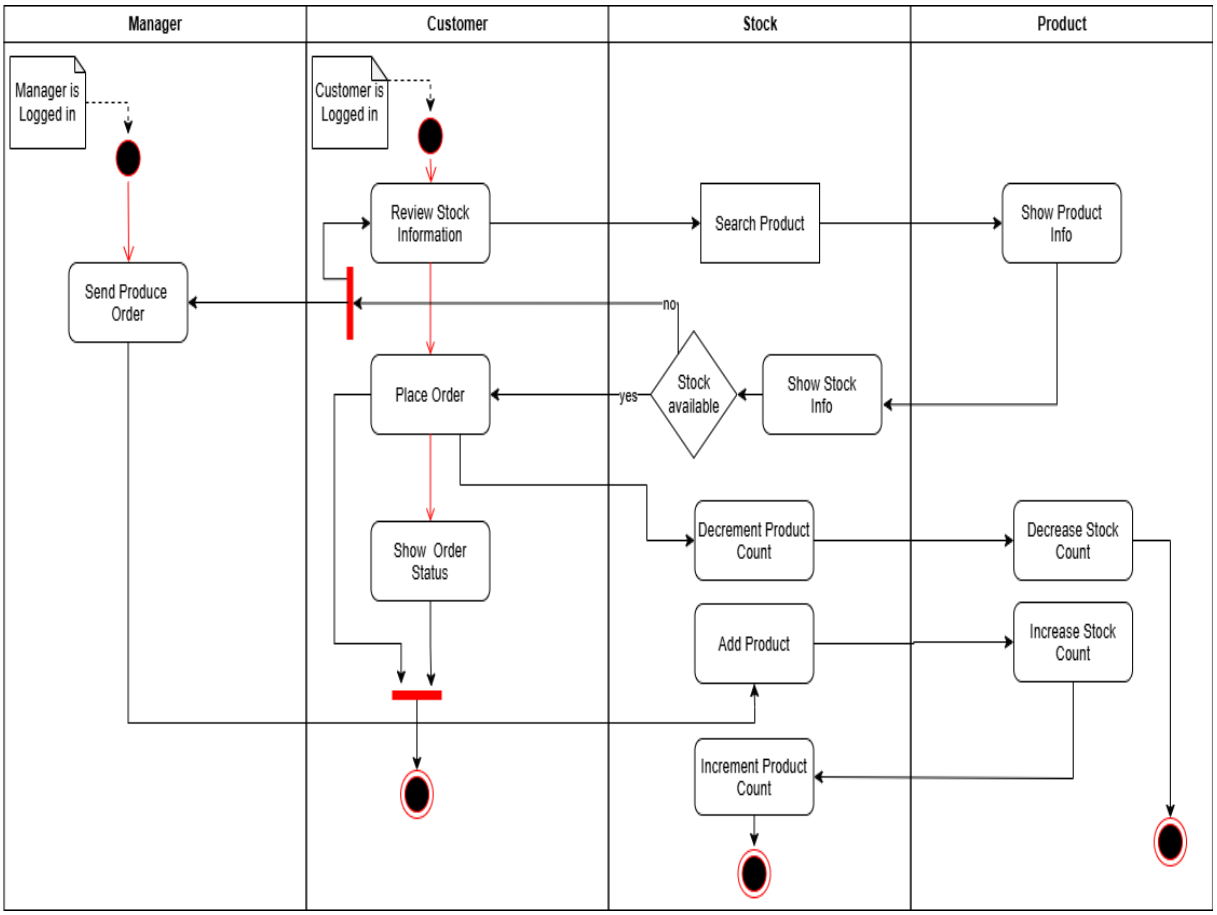
Statechart Diagram



Activity Diagram 1



Activity Diagram 2



Overall System Design Description

System Overview :

The Factory Management System is a role-based enterprise software to automate the process of factory operations. The system provides functionalities such as production monitoring, employee performance evolution, shift scheduling, stock controlling and customer order tracking. The main object is to improve effectiveness of the factory and reduce the manual effort.

User Roles and Functionalities :

Manager

- Monitoring production and employee performance
- See, approve, reject leave request
- Manage stocks, production, employee shifts
- Check payroll records

Employee

- View shift schedule and payroll
- Access insurance information
- Demand vacation
- See personal performance ratings

Customer

- Place and manage product orders
- Submit feedbacks on product experience
- See order status and review product availability

Design Principles

The system follows Object-Oriented Programming principles such as encapsulation, inheritance. This allows reusability of components, and easier debugging.